pfi

Python for Fixed Income

COURSE 8 · IMPLEMENTING AGENTIC AI

Structured Answers and Traced Numbers

WEEK 1 · DAY 4

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: structured answers and traced numbers

What day 3's saved answers did

2 of 6 wrote the tool's number as returned

4 of 6 rounded it, unasked

147 bp copied from a tool description

Today

1. Free text against a schema
2. Figure and DeskAnswer
3. A structured run
4. check_figures
5. A wrong digit, no tool behind it
6. numbers_untraced
7. Traces, and still wrong

Every model response in the notebook is a saved response, replayed. A live run can differ.

The rules for agents

What an agent produces: text, every number from a tool, now checked in code

Allowed names

the agent returned

the saved run

the brief states

Never called

the agent's opinion or view

what the agent believes

the agent's advice

On the synthetic book: a description, never a recommendation

Allowed names

DV01 per 100 of face

position DV01 in dollars

the largest DTS contributions in the file

Never called

cheap, rich, or attractive

a trade idea or a recommendation

overweight or underweight

Free text against a schema

The saved free-text answer to day 3's duration question

```
"As of 2026-09-30, the modified duration ... of bond 99126WAE0, ...

... at a 4.516% yield, is approximately 2.588 years."
```

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 2,238 prompt and 125 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

find it

the number, in a sentence

match it

to a tool result, by eye

count

the decimals it claims

Nothing in the sentence says which tool request a number came from.

How structured output works

The problem	A free-text answer cannot be checked by code without guessing
The schema	A pydantic model, turned by answer_format into a strict JSON schema
The constraint	Every field present, none added, each of the right type
You control	The fields, their types, and their descriptions: they are part of the prompt
What comes out	JSON that model_validate_json reads into a DeskAnswer, or an error

Figure and DeskAnswer

```
type: json_schema; name: DeskAnswer; strict: True
DeskAnswer: required ['text', 'figures', 'sources'], extra fields refused
 text string the answer in plain sentences
 figures array every number the text states
 sources array the files, sections, or statement dates used
Figure: required ['name', 'value', 'unit', 'tool_call_id'], extra fields refused
 name string what the number is, for example 'DV01 of the 2031 bond'
 value number the number, written to the decimals stated in the text
 unit string the unit as the tool result's key ends: per_100, pct, bp, usd, years
 tool_call_id string the id of the tool request whose result holds the number
```

answer_format: a strict JSON schema. Every field required, no extra field, descriptions sent as prompt.

How traced figures work

The problem	A number is worth what stands behind it
The id	Each tool result carries the id of its request, inside its JSON, since day 1
The copy	The schema asks for a tool_call_id with each figure; the model copies one
You control	The figure's fields and what their descriptions say
What comes out	Figures with a pointer into the run; nothing yet says the pointer is honest

A structured run

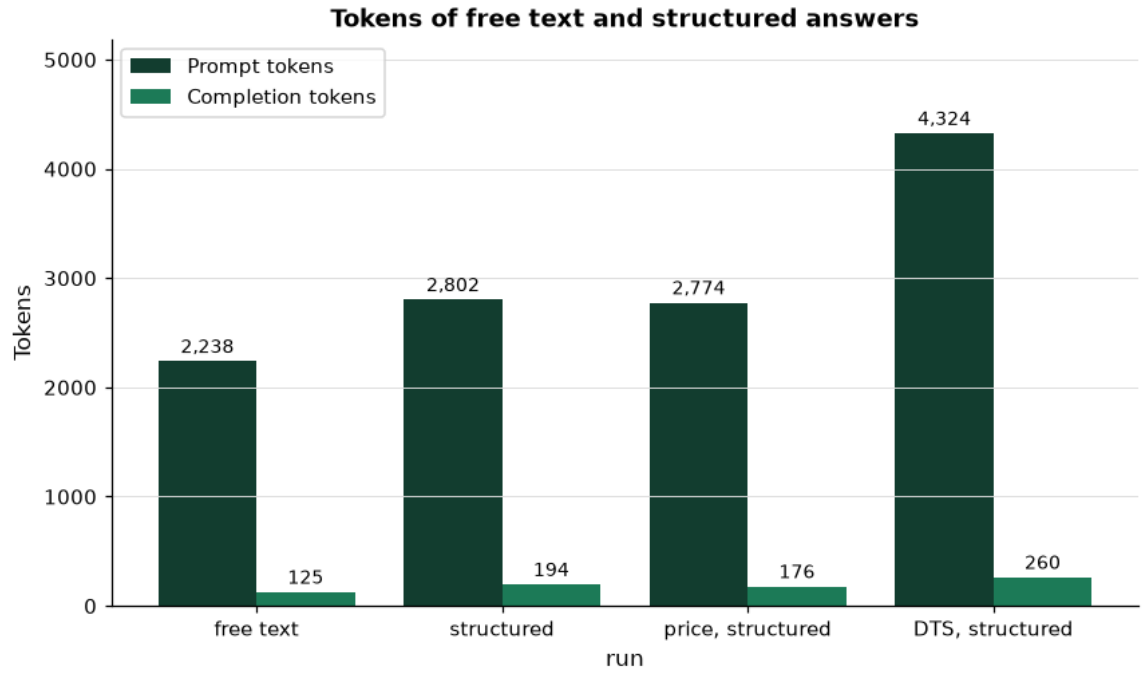
The same question with response_format=STRUCTURED: in the saved run

figure	`Modified duration of bond 99126WAE0: 2.587515 years`
cites	`call_ReR4zEhA1ezTVw3Wji8NBh5a`
run's ids	`'call_ReR4zEhA1ezTVw3Wji8NBh5a'`
text says	`"... approximately 2.588 years ..."`

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 2,802 prompt and 194 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

The figure carries the tool's number whole; the sentence still rounds it. Check both.

What a structured answer costs



Cost at the list price

free text	**US$0.001095**
structured	**US$0.001431**
price, structured	**US$0.001391**
DTS, structured	**US$0.002146**

The schema travels in the request and the figures are stated twice: a small price for an answer that can be checked.

How check_figures works

The problem	A pointer must be followed before it is trusted
The steps	Find the cited request; find a result number that rounds to the figure; check its key's unit
You control	The figure and the run; the rules are fixed
What comes out	A table, one row per figure: three checks and passed

check_figures: cited, at its rounding, in its unit

Three questions per figure

1. Is the cited request in this run?
2. Does its result, rounded to the figure's own decimals, equal the figure?
3. Does that number's key end in the unit?

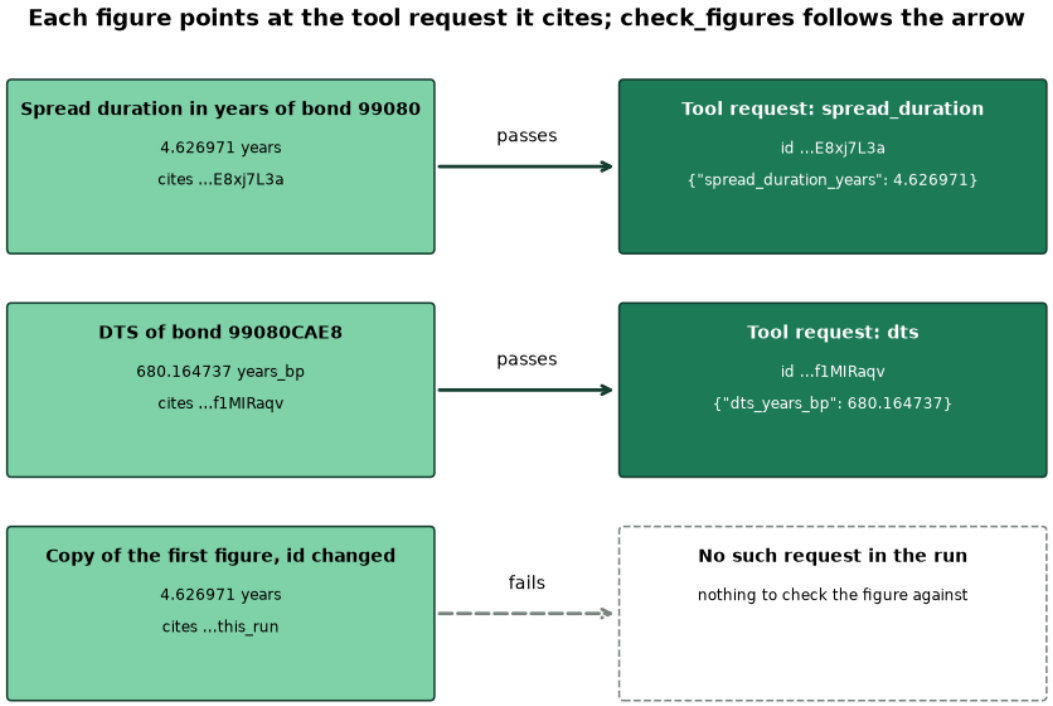
2.588 for 2.587515 passes; a wrong digit fails

In the saved runs

```
duration  1 of 1 passed
price     1 of 1 passed
dts       2 of 2 passed
```

No judgement and no reading of the text: an id, a number, a unit.

Each figure is a pointer



A figure is a pointer into the run, and the check follows the pointer.

A wrong digit, and no tool behind it

Two copies of the duration answer, checked against the same run

copy	value	cited	value_matches	passed
one digit changed	2.587516	True	False	False
no tool behind it	2.587515	False	False	False

What the agent did, step by step

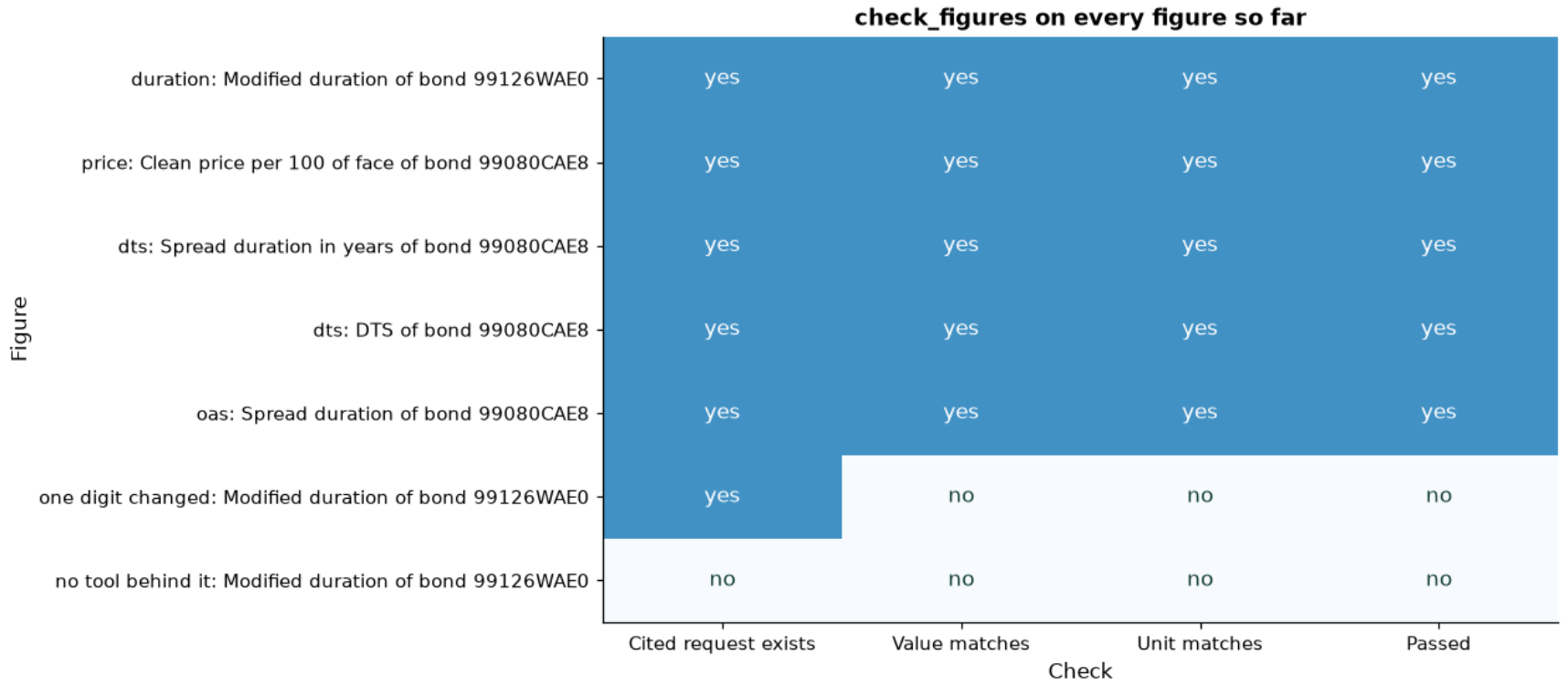
```
1  assistant dts  {"spread_duration_years":0,"spread_bp":0} tool_calls 1343 22

1  tool dts  {"error": "dts needs a spread duration and a spread, not 0.0 years and 0.0 bp...
```

The OAS question, structured: 2 dts requests with a 0 refused; one figure, the spread duration (figures: 1)

A figure with no tool behind it has nothing to be checked against, so it cannot pass.

check_figures on every figure so far



A check that cannot fail would show only dark cells. This one shows both.

How numbers_untraced works

The problem	Free text has no pointers, so the check is weaker
The skips	Dates, the phrase per 100, list numbers, and numbers the question wrote
The search	For every other number, a tool result of the run that holds it at its own decimals
You control	The text and the run
What comes out	The numbers no result holds; empty means traced, not right

numbers_untraced: checking free text

Every number in the text must trace, at its rounding, to some tool result

Skipped by rule: dates, ISO or in words

Skipped by rule: the unit phrase "per 100"

Skipped by rule: list ordinals ("1)", "2.")

Skipped by rule: every number the question wrote

```
untraced in the saved answer: []
untraced after one digit changed: ['2.589']
```

No citations, so no lookup: a number equal to some other result passes. check_figures is stricter.

What check_figures accepts

Unit	Spellings accepted	Count
`bp`	basis point, basis points, bps	3
`pct`	%, percent	2
`per_100`	per 100, per 100 of face	2
`usd`	$, dollars	2
`years`	year	1
`years_bp`	years times basis points, years times bp, years x bp	3

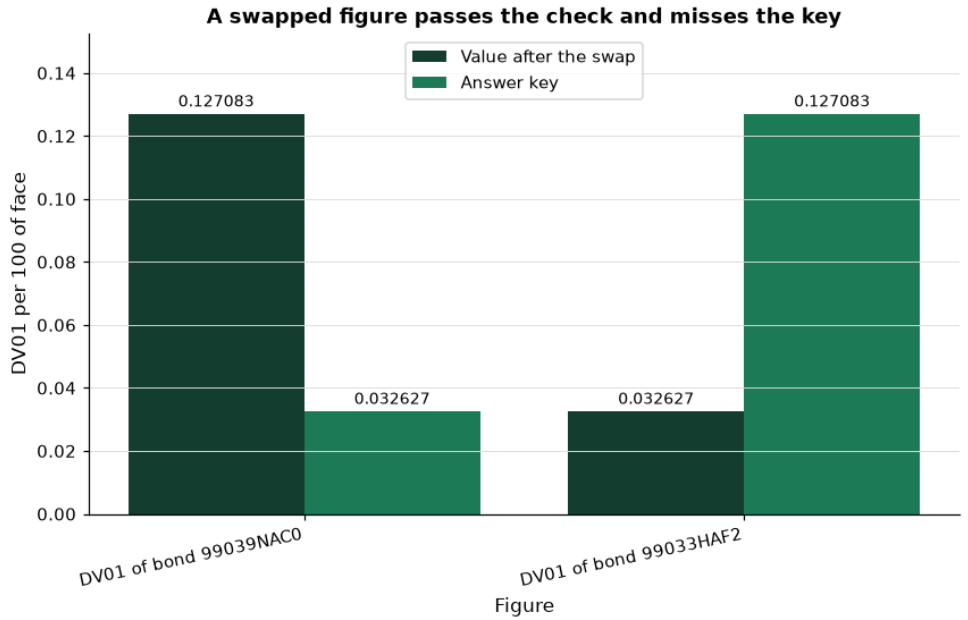
Tune first: nothing. These are rules of the check, not settings of a run.

Add a spelling only with a test: a rule widened to pass one answer passes the next wrong one.

How an answer key works

The problem	A traced figure can still answer a different question
The key	A value computed before any request, with the tool, on the bond the question names
The comparison	Each figure against its key, at the figure's decimals
You control	What the key covers: one value for each figure asked for
What comes out	An agreement or a difference the trace check never shows

A figure that traces can still be the wrong figure



Values and citations swapped

check_figures: 2 of 2 passed

Against the answer key: 2 of 2 differ

The names are kept; the values are swapped

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 3 requests, 4,487 prompt and 327 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

The right tool on the wrong bond passes every check here. Day 13 adds answer keys decided first.

Excel and Git

Excel side by side (tested in Excel)

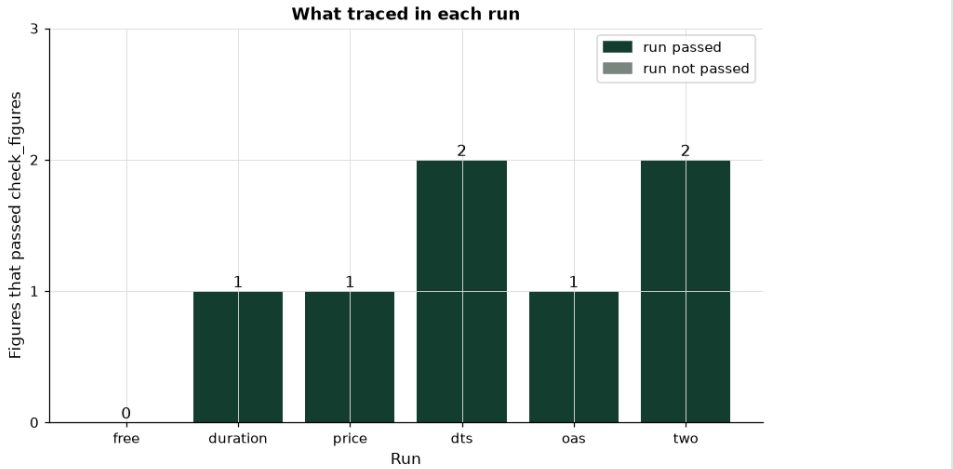
`=ROUND(B2,D2)=C2`	`B 2.587515  C 2.587515  D 6` (duration, figure)	TRUE
`=ROUND(B3,D3)=C3`	`B 2.587515  C 2.588  D 3` (duration, text)	TRUE
`=ROUND(B13,D13)=C13`	`B 2.587515  C 2.587516  D 6` (planted wrong digit, figure)	FALSE
`=ROUND(B15,2)=ROUND(C15,2)`	`0.0437 against 0.043625` (the AI check's bug)	TRUE
`=ROUND(B15,4)=C15`	`at the figure's own 4 decimals`	FALSE

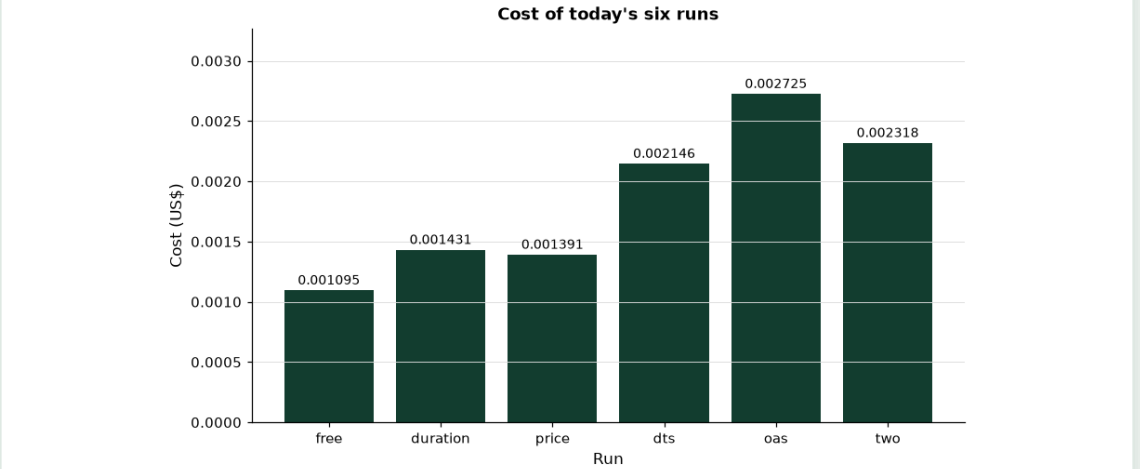
Excel against the course's checks: 12 of 12 rows the same

```
-free,gpt-4.1-mini-2025-04-14,2,2238,125,0.001095,2,answer,,
+free,gpt-4.1-mini-2025-04-14,2,2238,125,0.001095,2,answer,0,True
```

runs.csv: 6 rows, figures_traced and passed filled; git diff shows only those. 28 passed.

What traced in each run





Run	Requests	Prompt	Completion	Cost (US$)	Traced	Passed
free	2	2,238	125	0.001095	0	True
duration	2	2,802	194	0.001431	1	True
price	2	2,774	176	0.001391	1	True
dts	3	4,324	260	0.002146	2	True
oas	4	5,912	225	0.002725	1	True
two	3	4,487	327	0.002318	2	True

The AI check

The docstring is the prompt. The agent code that comes back runs and looks right.

1

Docstring

Round the tool value to the figure's own decimals. 0.0437 fails.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

0.0436 must match 0.043625 and 0.0437 must not. Run it: it fails.

→

4

Fix

Read the decimals from the figure; round only the tool value.

Today: both sides rounded to 2 decimals: 0.0437 and 0.043625 both become 0.04, so a wrong digit passes.

Code that runs is not the same as code that is right.

What you can do now

- Ask for an answer as a strict JSON schema
- Every figure names its tool request
- check_figures: cited, rounded, in its unit
- numbers_untraced for free text, exceptions stated
- A traced figure can still be the wrong bond

Week 1, Day 5 Case study: a risk agent on the synthetic book